HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2014)

Student No.: 02 Name: _____ Marks: / 44

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. $(2n^3)^{-5} =$

A. $\frac{1}{32n^2}$.

B. $\frac{1}{32n^{15}}$.

C. $\frac{1}{10n^{125}}$.

D. $\frac{1}{10n^{243}}$.

(DSE-2014 PII Q01)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

2. If p and q are constants such that $px(x-1)+x^2 = qx(x-2)+4x$, then $p =$

A. 1.

B. 2.

C. 3.

D. 4.

(DSE-2014 PII Q03)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

3. Let a be a constant. If the quadratic equation $x^2 + ax + a = 1$ has equal roots, then $a =$

A. -1.

B. 2.

C. 0 or -4.

D. 0 or 4.

(DSE-2014 PII Q04)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

4. If $a > b$ and $k < 0$, which of the following must be true?

I. $a^2 > b^2$

II. $a + k > b + k$

III. $\frac{a}{k^2} > \frac{b}{k^2}$

A. I only

B. II only

C. I and III only

D. II and III only

(DSE-2014 PII Q06)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

5. The solution of $-3x < 6 < 2x$ is

A. $x > -2$.

B. $x > 0$.

C. $x > 3$.

D. $-2 < x < 3$.

(DSE-2014 PII Q07)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

6. There are 792 workers in a factory. If the number of male workers is 20% less than that of female workers, then the number of male workers is

A. 352.

B. 360.

C. 432.

D. 440.

(DSE-2014 PII Q09)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

7. The width and the length of a thin rectangular metal sheet are measured as 8 cm and 10 cm correct to the nearest cm respectively. Let x cm^2 be the actual area of the metal sheet. Find the range of values of x.

A. $71.25 \leq x < 89.25$

B. $71.25 < x \leq 89.25$

C. $79.5 \leq x < 80.5$

D. $79.5 < x \leq 80.5$

(DSE-2014 PII Q11)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

8. Let a_n be the nth term of a sequence. If $a_2 = 7$, $a_4 = 63$ and $a_{n+2} = a_{n+1} + a_n$ for any positive integer n, then $a_5 =$

A. 56.

B. 70.

C. 91.

D. 119.

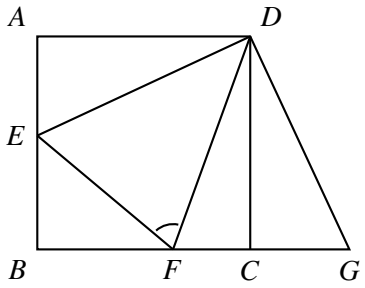
(DSE-2014 PII Q14)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

9. In the figure, $ABCD$ is a square. BC is produced to G such that $\angle CDG = 25°$. E is a point lying on AB such that $AE = CG$. If F is a point lying on BC such that $\angle CDF = 20°$, then $\angle DFE =$



A. 60°.

B. 65°.

C. 70°.

D. 73°.

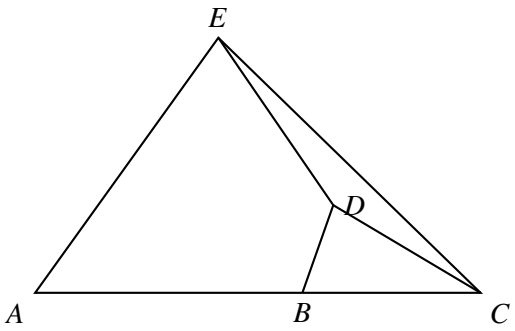
(DSE-2014 PII Q16)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

10. In the figure, B is a point lying on AC such that $AB : BC = 3 : 2$. It is given that $AE \parallel BD$. If the area of $\triangle BCD$ and the area of $\triangle CDE$ are 4 cm^2 and 8 cm^2 respectively, then the area of the trapezium $ABDE$ is



A. 18 cm^2.

B. 21 cm^2.

C. 27 cm^2.

D. 33 cm^2.

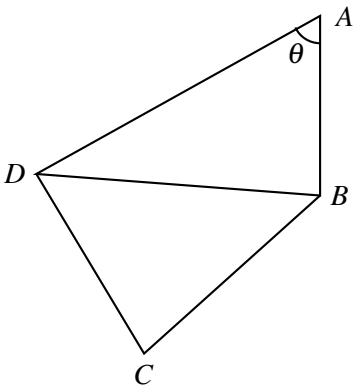
(DSE-2014 PII Q17)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

11. In the figure, $\angle ABD = \angle ADC = \angle BCD = 90°$. If $AB = \ell$, then $CD =$



A. $\ell \sin\theta$.

B. $\ell \cos\theta$.

C. $\ell \sin\theta \tan\theta$.

D. $\dfrac{\ell \tan\theta}{\cos\theta}$.

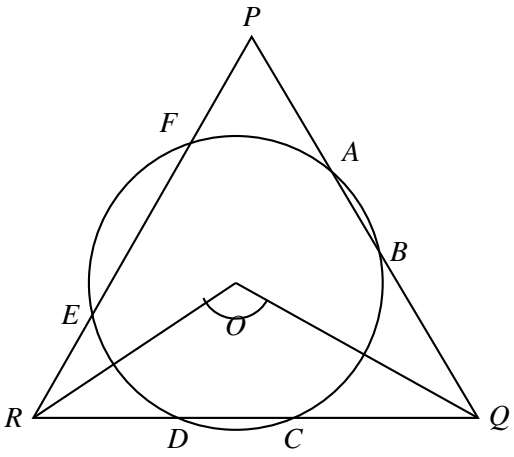
(DSE-2014 PII Q18)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

12. In the figure, O is the centre of the circle $ABCDEF$. $\triangle PQR$ intersects the circle at A, B, C, D, E and F. If $\angle QPR = 38^\circ$ and $AB = CD = EF$, then $\angle QOR =$



A. 109°.

B. 117°.

C. 123°.

D. 142°.

(DSE-2014 PII Q21)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

13. The rectangular coordinates of the point P are $(-1, \sqrt{3})$. If P is reflected with respect to the x-axis, then the polar coordinates of its image are

A. $(2, 210°)$.

B. $(2, 240°)$.

C. $(4, 210°)$.

D. $(4, 240°)$.

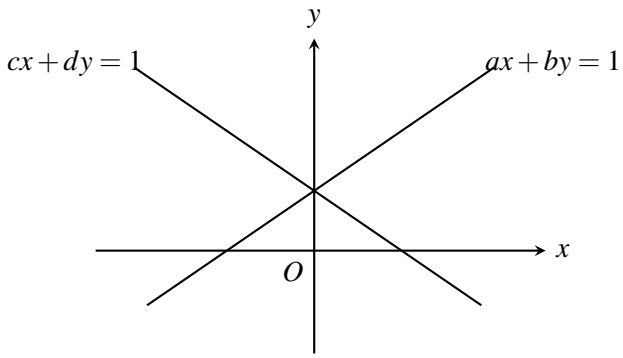
(DSE-2014 PII Q23)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

14. In the figure, the two straight lines intersect at a point on the positive y-axis. Which of the following are true?



I. $a < 0$

II. $c > 0$

III. $b = d$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2014 PII Q25)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

15. If a diameter of the circle $x^2 + y^2 - 8x + ky - 214 = 0$ passes through the point $(6, -5)$ and the slope of the diameter is -4, then $k =$

A. -6.

B. -4.

C. 13.

D. 70.

(DSE-2014 PII Q26)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

16. The stem-and-leaf diagram below shows the distribution of the ages of the passengers in a bus.

Stem (tens)	Leaf (units)
1	h 4 6
2	3 3 3 4 6 7 7
3	1 2 2 2 6 8
4	0 k

If the range of the above distribution is at least 33, which of the following must be true?

I. $0 \leq h \leq 3$

II. $3 \leq k \leq 9$

III. $3 \leq k - h \leq 5$

A. I only

B. II only

C. I and III only

D. II and III only

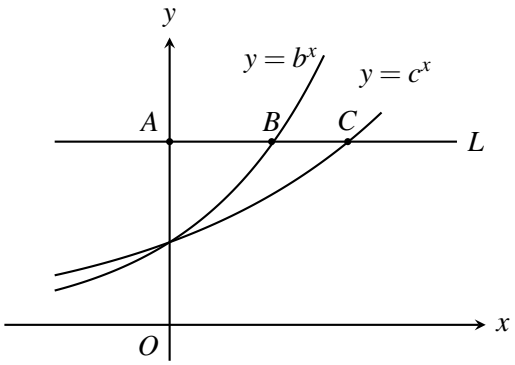
(DSE-2014 PII Q30)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

17. The figure shows the graph of $y = b^x$ and the graph of $y = c^x$ on the same rectangular coordinate system, where b and c are positive constants. If a horizontal line L cuts the y-axis, the graph of $y = b^x$ and the graph of $y = c^x$ at A, B and C respectively, which of the following are true?



I. $b < c$

II. $bc > 1$

III. $\dfrac{AB}{AC} = \log_b c$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2014 PII Q32)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

18. Which of the following is the greatest?

A. 124^{241}

B. 241^{214}

C. 412^{142}

D. 421^{124}

(DSE-2014 PII Q33)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

19. $7 \times 2^{10} + 2^8 + 5 \times 2^3 - 2^3 =$

A. 111010100000_2.

B. 111100010000_2.

C. 1110100100000_2.

D. 1111000010000_2.

(DSE-2014 PII Q34)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

20. Let $f(x) = 3x^2 - 6x + k$, where k is a constant. If the y-coordinate of the vertex of the graph of $y = f(x)$ is 7, then $k =$

A. 1.

B. 3.

C. 4.

D. 10.

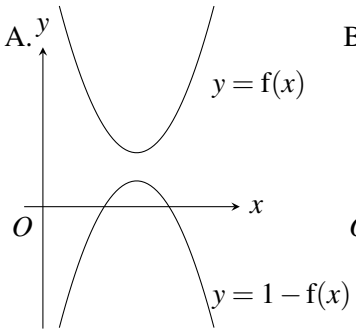
(DSE-2014 PII Q35)

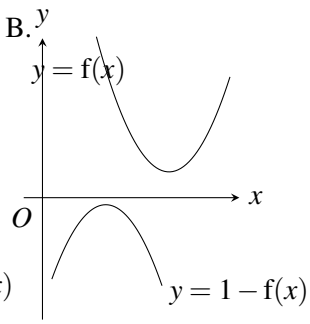
Answers written in the margins will not be marked.

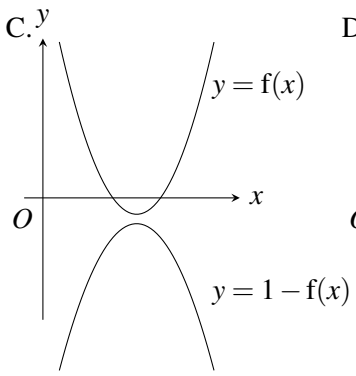
Answers written in the margins will not be marked.

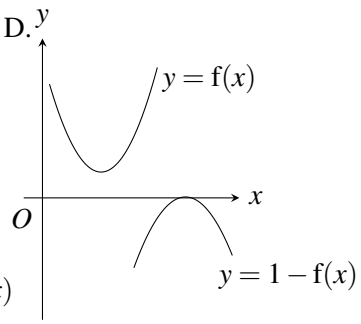
Answers written in the margins will not be marked.

21. Which of the following may represent the graph of $y = \mathrm{f}(x)$ and the graph of $y = 1 - \mathrm{f}(x)$ on the same rectangular coordinate system?

A.


B.


C.


D.


(DSE-2014 PII Q38)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

22. For $0° \leq x \leq 360°$, how many roots does the equation $7\sin^2 x = \sin x$ have?

A. 2

B. 3

C. 4

D. 5

(DSE-2014 PII Q39)

I can finish _____ questions

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.